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Embedded AI Project Report

Efficient Vision Transformers for Embedded Anomaly Detection

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Abstract

This project presents an embedded anomaly detection pipeline built around Vision Transformers (ViTs) and a set of model-compression techniques intended for resource-constrained deployment. The primary accuracy metric used throughout this study is the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), which provides a threshold-independent measure of model discrimination ability and is well-suited for evaluating anomaly detection systems. The study begins with a ViT-B/16 baseline and evaluates the effect of pruning, token reduction, quantization, and knowledge distillation on accuracy (measured via AUC-ROC), latency, and model size. The experiments show that aggressive compression can significantly reduce model footprint, but the gains are not always accompanied by improvements in predictive performance. In particular, several lightweight configurations preserve the baseline AUC-ROC, while some structural reduction methods improve AUC-ROC, and others degrade it. The resulting analysis provides a practical view of the speed–accuracy trade-offs involved in adapting transformer-based anomaly detection to embedded environments.

1. Introduction

Anomaly detection is widely used in industrial inspection, smart monitoring, surveillance, and healthcare analytics. In such settings, models must not only be accurate but also efficient enough for real-time execution on edge or embedded hardware.

Vision Transformers have emerged as a strong alternative to conventional convolution-based backbones because they model global dependencies through self-attention. However, this expressiveness comes at the cost of a larger compute footprint and higher memory demand. For embedded deployment, a practical solution often requires compression techniques such as quantization, pruning, distillation, and token reduction.

The goal of this project is to analyze how these techniques affect anomaly-detection performance when applied to a ViT-based system. Since anomaly detection is inherently a binary classification problem with potentially imbalanced classes, we adopt the AUC-ROC (Area Under the ROC Curve) as the primary accuracy metric. AUC-ROC measures the model's ability to discriminate between normal and anomalous samples across all decision thresholds, making it a more informative accuracy indicator than simple classification accuracy in this setting. The experiments focus on a baseline model and a sequence of optimization strategies, with performance measured using AUC-ROC, EER, latency, memory usage, and model size.

2. Objectives

The project objectives are:

- build a ViT-based anomaly detection baseline,
- evaluate model compression methods for embedded deployment,

- compare accuracy (measured via AUC-ROC) and latency across optimization strategies,
- quantify the effect of each technique on model size,
- identify configurations that provide a better efficiency–accuracy trade-off.

3. Methodology

3.1. Pipeline Overview

The system follows a standard embedded AI workflow:

1. image acquisition and preprocessing,
2. patch embedding and transformer encoding,
3. anomaly scoring using a classification or reconstruction-based objective,
4. evaluation under baseline and compressed configurations.

3.2. Baseline Model

The baseline uses a ViT-B/16 configuration and serves as the reference point for all experiments. The baseline performance reported in the results summary is:

$$\text{AUC-ROC} = 0.4778, \quad \text{Latency} = 10.41 \text{ ms}, \quad \text{Size} = 349.84 \text{ MB}.$$

3.3. Compression Techniques

The following techniques were evaluated:

- **Pruning:** unstructured L1 pruning, iterative L1 pruning, head pruning, and FFN channel pruning,
- **Token Reduction:** ToMe-style token merging and DynamicViT-style token keeping,
- **Quantization:** FP16 and INT8 post-training quantization,
- **Knowledge Distillation:** teacher-student transfer from ViT-B/16 to DeiT-Tiny.

3.4. Evaluation Metrics

Model performance is assessed using the following metrics:

- **AUC-ROC (Primary Accuracy Metric):** The Area Under the Receiver Operating Characteristic Curve is used as the main accuracy measure throughout this project. It quantifies a model's ability to discriminate between normal and anomalous samples in a threshold-independent manner. A value of 1.0 represents perfect classification, while 0.5 indicates random chance. AUC-ROC is preferred over standard classification accuracy for anomaly detection tasks because it is robust to class imbalance and captures performance across all operating points simultaneously.

- **EER:** the Equal Error Rate, which is the point at which the false acceptance rate equals the false rejection rate; lower values indicate better performance.
- **Latency:** inference time per frame in milliseconds, reflecting real-time deployment feasibility.
- **Memory / Size:** resource usage for deployment comparison across compressed configurations.

4. Baseline Results

Figure 1 summarizes the baseline training curve, ROC curve, and anomaly-score distribution. The model converges smoothly, and the score distributions show partial overlap between normal and anomalous samples, which explains the moderate AUC-ROC value. Here, AUC-ROC serves as the accuracy metric indicating how well the model separates anomalous from normal samples.

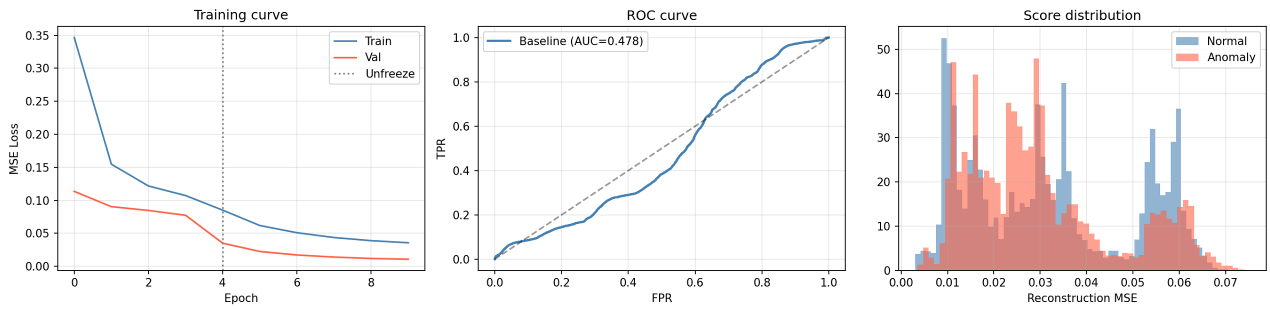

Figure 1: Baseline training, ROC curve, and score distribution results. AUC-ROC is used as the primary accuracy metric.

Table 1: Baseline reference metrics. AUC-ROC is the accuracy metric used for model evaluation.

Model	AUC-ROC (Accuracy Metric)	Latency (ms)	Size (MB)
ViT-B/16 baseline	0.4778	10.41	349.84

5. Pruning Experiments

Figure 2 compares pruning strategies in terms of AUC-ROC (the accuracy metric) and latency. Unstructured L1 pruning shows a small improvement at higher sparsity, while iterative L1 pruning gives a modest gain only at the larger sparsity setting. Head pruning with Taylor scores provides the strongest increase in AUC-ROC among the pruning experiments. FFN channel pruning is more mixed and does not consistently outperform the baseline.

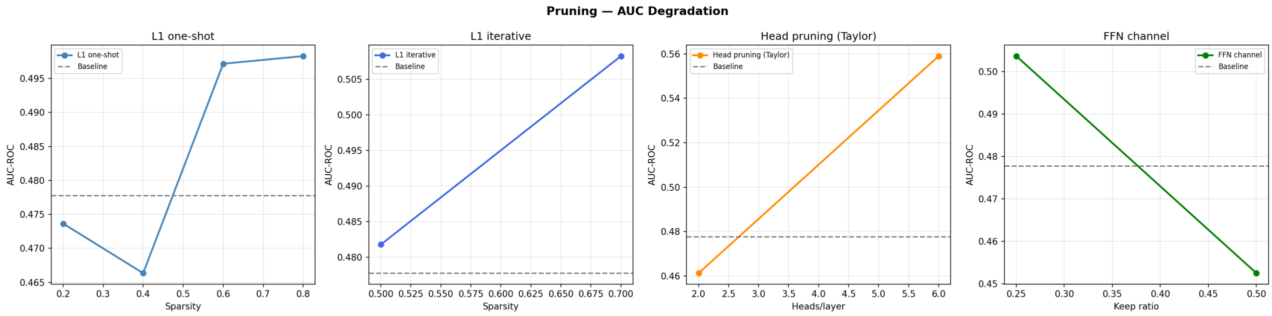

Figure 2: AUC-ROC (accuracy metric) degradation and gain trends under different pruning strategies.

Table 2: Pruning summary. AUC-ROC is the accuracy metric used for comparison across all pruning configurations.

Method	Parameter	AUC-ROC (Accuracy)	EER	Latency (ms)	Size (MB)
L1 one-shot	0.2	0.4736	0.5521	10.36	349.84
L1 one-shot	0.4	0.4663	0.5262	10.18	349.84
L1 one-shot	0.6	0.4972	0.4789	10.11	349.84
L1 one-shot	0.8	0.4983	0.4777	10.06	349.84
L1 iterative	0.5	0.4818	0.5268	10.30	349.84
L1 iterative	0.7	0.5083	0.4814	10.33	349.84
Head pruning (Taylor)	2 heads/layer	0.4612	0.5667	10.26	349.84
Head pruning (Taylor)	6 heads/layer	0.5590	0.4453	10.12	349.84
FFN channel	0.50 keep ratio	0.4525	0.5693	7.85	236.52
FFN channel	0.25 keep ratio	0.5037	0.5234	7.07	179.86

6. Token Reduction

Figure 3 compares token reduction methods in terms of accuracy (AUC-ROC) and latency. ToMe achieves the best AUC-ROC at merge ratio 4, but the accuracy drops as merging becomes more aggressive. DynamicViT improves over the baseline AUC-ROC at a keep ratio of 0.7, but too much token dropping causes degradation.

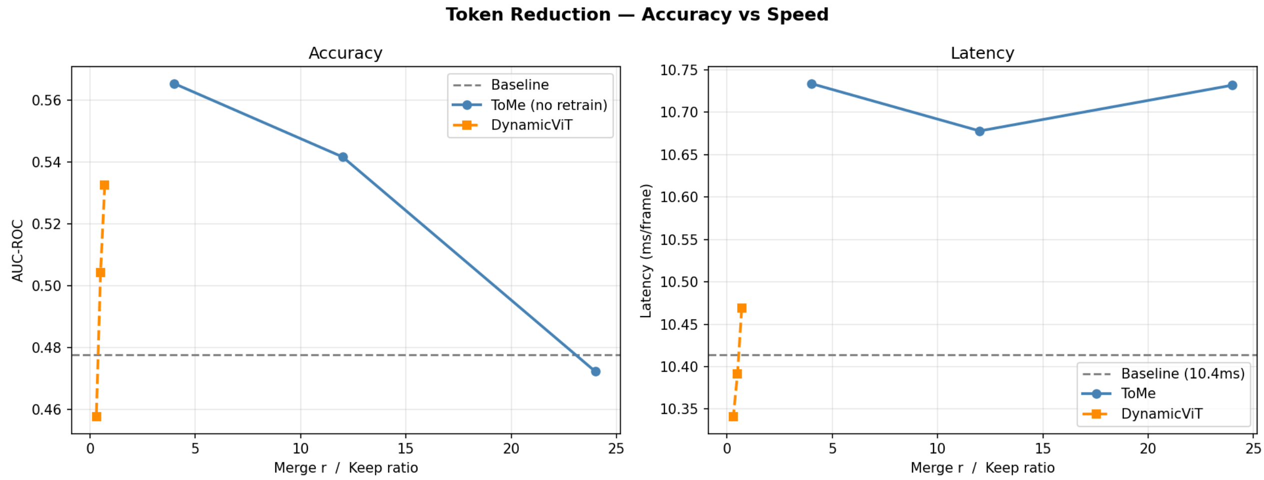

Figure 3: Accuracy (AUC-ROC) and latency trade-off under token reduction strategies.

Table 3: Token reduction results. AUC-ROC is the accuracy metric used for evaluation.

Method	Parameter	AUC-ROC (Accuracy)	EER	Latency (ms)
ToMe	merge $r = 4$	0.5653	0.4605	10.73
ToMe	merge $r = 12$	0.5416	0.4771	10.68
ToMe	merge $r = 24$	0.4723	0.5469	10.73
DynamicViT	keep 0.7	0.5325	0.4937	10.47
DynamicViT	keep 0.5	0.5042	0.5012	10.39
DynamicViT	keep 0.3	0.4577	0.5125	10.34

7. Quantization Results

Figure 4 summarizes the quantization study in terms of accuracy (AUC-ROC), latency, and model size. FP16 preserves the baseline AUC-ROC while halving the model size and improving latency. The INT8 dynamic PTQ configuration greatly reduces size but introduces a large latency penalty in the recorded experiment, indicating that implementation details and backend support strongly affect real deployment outcomes.

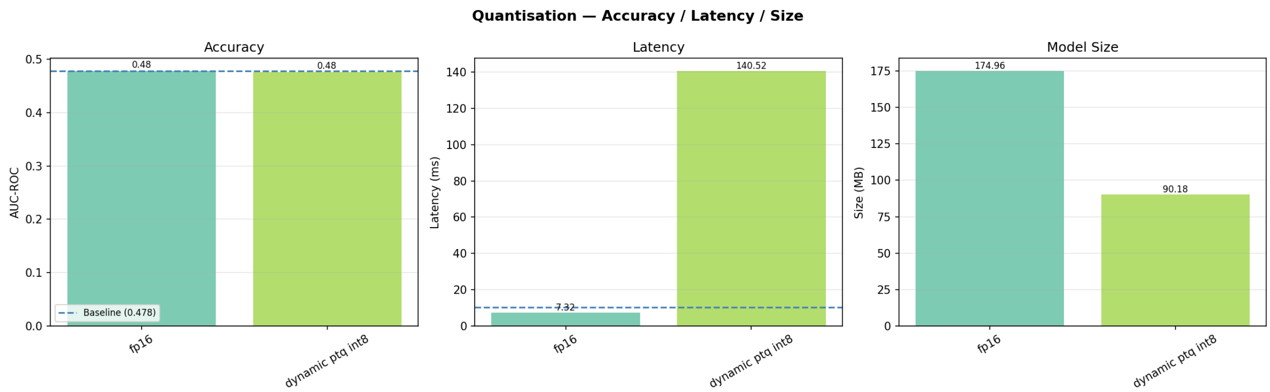

Figure 4: Quantization comparison across accuracy (AUC-ROC), latency, and model size.

Table 4: Quantization results. AUC-ROC is the accuracy metric used to assess the impact of quantization on model performance.

Mode	AUC-ROC (Accuracy)	Latency (ms)	Size (MB)
FP16	0.4777	7.32	174.96
Dynamic PTQ INT8	0.4765	140.52	90.18

8. Knowledge Distillation

Figure 5 shows the training behavior of the DeiT-Tiny student distilled from the ViT-B/16 teacher. The student matches the teacher's AUC-ROC (accuracy metric) while reducing model size substantially and lowering latency, demonstrating that knowledge distillation is an effective strategy for preserving accuracy under aggressive compression.

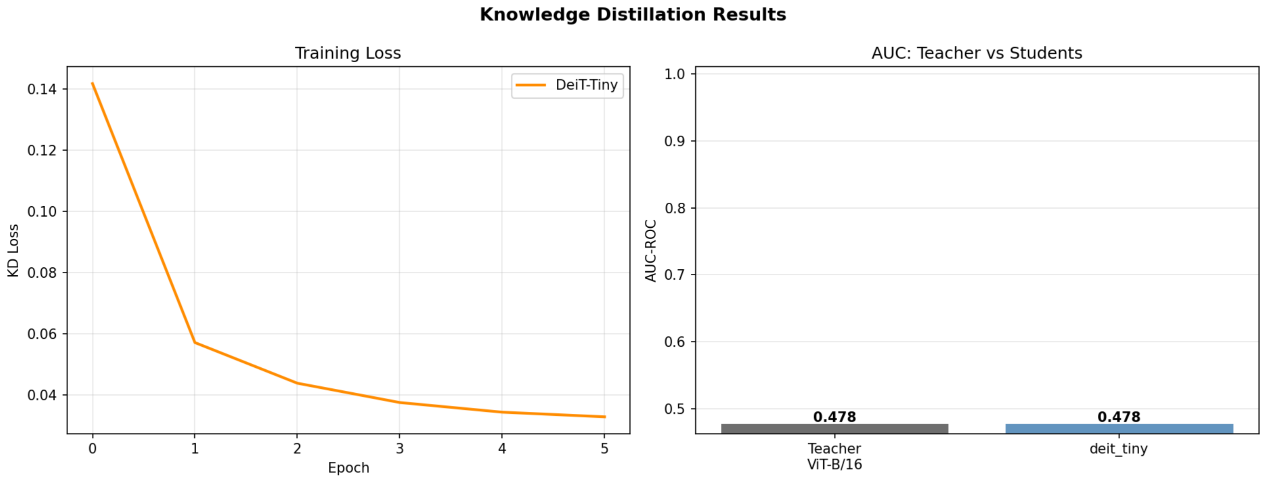

Figure 5: Knowledge distillation training loss and teacher–student comparison. AUC-ROC is used as the accuracy metric for both models.

Table 5: Knowledge distillation summary. AUC-ROC is the accuracy metric showing the student model matches the teacher.

Model	AUC-ROC (Accuracy)	Latency (ms)	Size (MB)
Teacher ViT-B/16	0.4777	10.41	349.84
DeiT-Tiny student	0.4777	6.95	25.35

9. Discussion

Across all experiments, accuracy is measured using AUC-ROC, which was chosen as the primary accuracy metric because anomaly detection involves binary classification under potential class imbalance, where AUC-ROC provides a more reliable measure of model performance than standard accuracy. The results show that compression does not produce a single universal best setting. Some techniques primarily improve deployment efficiency, while others improve accuracy (AUC-ROC) or both.

The most notable observations are:

- **Best accuracy (AUC-ROC) among pruning variants:** head pruning with 6 heads/layer, AUC-ROC = 0.5590.
- **Best accuracy (AUC-ROC) among token reduction methods:** ToMe with merge $r = 4$, AUC-ROC = 0.5653.
- **Best latency among the reported methods:** DeiT-Tiny student, 6.95 ms.
- **Largest size reduction:** DeiT-Tiny student, 25.35 MB.
- **Best practical quantized baseline:** FP16, because it preserves accuracy (AUC-ROC) while reducing size and latency.

These findings suggest that token reduction and distillation are more promising than naive aggressive quantization for this specific implementation. In contrast, pruning can help, but its benefit (in terms of AUC-ROC accuracy) depends strongly on the pruning granularity and parameter choice.

10. Conclusion

This project demonstrates a full embedded AI optimization study for anomaly detection using Vision Transformers. AUC-ROC is adopted as the primary accuracy metric throughout, as it provides a threshold-independent measure of how well the model distinguishes normal from anomalous samples. Starting from a ViT-B/16 baseline, the work evaluates pruning, token reduction, quantization, and knowledge distillation under real measured metrics. The experiments show that careful compression can reduce latency and model size without necessarily harming accuracy (as reflected in AUC-ROC), but overly aggressive reduction can quickly degrade performance.

Overall, the results indicate that the best deployment strategy depends on the target constraint. For accuracy preservation (as measured by AUC-ROC), FP16 and selected pruning/token-reduction settings are effective. For strongest compression, knowledge distillation delivers the smallest and fastest model while maintaining the same AUC-ROC as the original teacher.

11. Future Work

Future extensions of this project may include:

- evaluating more lightweight transformer backbones,
- combining distillation with quantization-aware training,
- testing the system on additional anomaly-detection datasets,

- deploying and benchmarking on actual embedded hardware,
- improving the dynamic quantization backend to remove latency overhead.

References

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